

EXERCISE 5: NULL FUNCTIONS

PROBLEM ZINDOYE

Q1 Employee salaries

```
SELECT employee-id,  
       name,  
       IFNULL(salary, 0) AS salary-with-default  
FROM employees;
```

Output

employee-id	name	salary-with default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2.

```
SELECT employee-name,  
       name,  
       IFNULL(department, 'Not Assigned') AS department-name  
FROM employees;
```

Output

employee-id	name	department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Q3

```
SELECT *  
FROM orders  
WHERE customer IS NULL;
```

Output

order_id	customer_id	delivery_date
103	NULL	2024-12-03

Q4

```
SELECT order_id,  
       customer_id,  
       delivery_date,  
       IFNULL(CAST(delivery_date AS STRING), 'Pending') AS  
       delivery_status  
FROM orders;
```

Output

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Q5

```
SELECT student_id,  
       name,  
       IFNULL(grade, 0) AS final_grade  
FROM students;
```

Q5 continued

Output

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6

```
SELECT COUNT(*) AS students_not_graded
FROM students
WHERE grade IS NULL
```

Output

students_not_graded
1

Q7

```
SELECT product_id,
       name,
       price-IFNULL(discount, 0) AS final_price
FROM products;
```

Output

name	price	final-price
Keyboard	25	25
Mouse	15	10
Monitor	100	100

Q8

```
SELECT DISTINCT (*) AS missing_email_count  
FROM customers  
WHERE email IS NULL
```

Output

missing_email_count
2

Q9

```
SELECT customer_id,  
       name,  
       IFNULL(email, 'No Email') AS email_display  
FROM customers;
```

Output

customer id	name	email_display
1	Linda	No Email
2	Joseph	joseph@gmail.com
3	Nia	No Email

Q10

```
SELECT payment_id,  
       IFNULL(method, 'unknown') AS method_display,  
       status,  
FROM payments;
```

Output

payment_id	method	status
301	Credit	NULL
302	Paypal	Success
303	Unknown	Failed

Q11

```
SELECT item_id,  
       item_name,  
       IFNULL(quantity, 0) AS quantity_checked  
FROM inventory;
```

Output

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Q12

```
SELECT emp_id,  
       COALESCE(bonus, commission, 0000) AS first_available_reward  
FROM employees_extra;
```

Output

emp_id	first_available_reward
1	300
2	100
3	0000

Q13

```
SELECT COUNT(*) AS no_room-count  
FROM classes  
WHERE room IS NULL;
```

Q13 continued

Output

student_id	date	status
1	2025-04-01	NULL
2	2025-04-01	Present
3	2025-04-01	Absent

Q14

```
SELECT student_id,  
       date  
       IFNULL(status, 'Not Marked') AS attendance_status  
FROM attendance;
```

Output

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Q15

```
SELECT account_id,  
       IFNULL(account_type, 'Unknown') AS type_display,  
       IFNULL(balance, 0) AS balance_checked  
FROM bank_accounts;
```

Output

account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Q16

```
SELECT project_id,  
       title,  
       COALESCE (CAST (start-date AS STRING, 'TBD') AS start-displaystart-date)  
FROM   projects;
```

Output

project_id	title	start-date	start-display
1	Website Revamp		2025-01-10
2	Mobile App		TBD
3	Data Migration		TBD

Q17

```
SELECT review_id,  
       product_id,  
       IFNULL (comment, 'No Comment') AS comment-display,  
       IFNULL (rating, 0) AS rating-display  
FROM   reviews;
```

Output

review_id	product_id	comment-display	rating-display
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3

Q18

```
SELECT supplier_id,  
       name,  
       COALESCE (phone, alt-phone, 0000) AS contact-number  
FROM   suppliers;
```

Q18 Continued

supplier-id	name	contact number
1	Global Goods	1 2 3 4 5 6 7 8 9
2	Bet Supplies	9 8 7 6 5 4 3 2 1
3	Value Source	0 0 0 0

Q19

```

SELECT user_id,
       COALESCE (theme, 'Light') AS theme_set,
       COALESCE (language, 'English') AS language_set,
       COALESCE (timezone, 'UTC') AS timezone_set
FROM   user_settings;

```

Output

user_id	theme	language	timezone
1	Light	English	UTC UTC
2	Dark	English	UTC+1
3	Light	English	UTC

Q20

```

SELECT record_id,
       machine_id,
       COALESCE (issue, 'Unknown Issue') AS issue_log,
       COALESCE (technician, 'Not Assigned') AS technician_name
FROM   maintenance;

```

Output

record_id	machine_id	issue_log	technician
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex